

**Objective: Open Ended Lab**

**PROBLEM STATEMENT:**

Design and Implement a Database Management System for a Real-World Application.

**PROBLEM DESCRIPTION:**

In this laboratory assignment, you are tasked with designing and implementing a Database Management System (DBMS) for a real-world application of your choice. The goal is to demonstrate your understanding of database design principles, schema normalization, query optimization, and practical implementation of a DBMS.

**INSTRUCTIONS:**

1. Select a real-world application that requires a database system. Examples may include an online e-commerce platform, a library management system, a hospital patient information system, or a social networking platform. Ensure that the chosen application is sufficiently complex to showcase your skills in database design and implementation.
2. Design the database schema for the selected application. Start by identifying the entities, attributes, and relationships involved. Use appropriate entity-relationship (ER) modeling techniques to represent the database schema.
3. Normalize the database schema to ensure that it is in an optimal form, minimizing data redundancy and ensuring data integrity. Apply normalization techniques such as First Normal Form (1NF), Second Normal Form (2NF), and Third Normal Form (3NF) as applicable.
4. Implement the database schema using a relational DBMS. Create the necessary tables, define appropriate data types and constraints, and establish the relationships between tables.
5. Populate the database with representative data to simulate real-world scenarios. Ensure that the data reflects various aspects of the selected application, such as different user profiles, transactions, or entities.
6. Develop SQL queries that showcase the functionality of the DBMS. Include queries for data retrieval, data manipulation, and complex operations that highlight the capabilities of the DBMS and demonstrate your proficiency in SQL.
7. Optimize the query performance by analyzing query execution plans, creating appropriate indexes, and utilizing database optimization techniques such as query rewriting, de-normalization, or materialized views.
8. Test and validate the database system by executing the implemented queries and performing various operations on the database. Verify that the system functions correctly, handles concurrency, and maintains data consistency.
9. Document your database design, implementation decisions, and any challenges faced during the process. Provide a detailed explanation of the schema, relationships, constraints, and optimizations performed.
10. Present your database system, highlighting its features, performance, and usability.